Lab 08

Creating an API Project using the API Controller

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs and paths in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal, Admin Portal, Management Console, token endpoint).
- **`<GW_HTTPS_PORT>` / `<GW_HTTP_PORT>`** – API Gateway ports: 8243 (HTTPS) and 8280 (HTTP) plus any port offset.
- **`<APIM_HOME>`** – the folder where API Manager is extracted.
- **`<APICTL_DIR>`** – the folder that contains the apictl program after you extract it, for example ~/apictl (Windows: C:\apictl).

The servers use a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue. The -k option in the apictl commands skips the certificate check for the same reason.

Training Objective

Learn how to create an API and publish it to the Developer Portal using the WSO2 API Controller (apictl).

Business Scenario

After setting up API Manager, an API is usually created and published through the Publisher portal so that it can be subscribed to from the Developer Portal.

In this exercise, an API developer has an OpenAPI definition for a REST service and needs to deploy it in API Manager, and develop and manage the API, without using the Publisher portal.

High-Level Steps

- Install and initialise apictl
- Add the API Manager environment and log in
- Create the API project
- Publish the API

Part 1 – Install and Initialise apictl

1. Download the API Controller (apictl) 4.7.x for your operating system from the WSO2 API Manager **Tooling** download page, and extract it. You get a folder named `apictl` that contains the `apictl` program (Windows: `apictl.exe`).

macOS / Linux (for example darwin-arm64 for an Apple silicon Mac)

```
tar -xzf apictl-4.7.<x>-<os>-<arch>.tar.gz -C ~/
```

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On Windows, right-click the downloaded zip, select **Extract All** and extract it to `C:\`. This gives `C:\apictl`.

2. Add the apictl folder to your PATH so that the `apictl` command works from any folder. Without this step the terminal shows `command not found: apictl`.

   macOS (zsh)

   ```
   echo 'export PATH="$PATH:<APICTL_DIR>"' >> ~/.zshrc
   source ~/.zshrc
   ```

   Example – macOS (zsh):

   ```
   echo 'export PATH="$PATH:~/apictl"' >> ~/.zshrc
   source ~/.zshrc
   ```

   Linux (bash)

   ```
   echo 'export PATH="$PATH:<APICTL_DIR>"' >> ~/.bashrc
   source ~/.bashrc
   ```

   Example – Linux (bash):

   ```
   echo 'export PATH="$PATH:~/apictl"' >> ~/.bashrc
   source ~/.bashrc
   ```

   Windows (Command Prompt) – then close it and open a new Command Prompt

   ```
   setx PATH "%PATH%;C:\apictl"
   ```

   On macOS, if a security message blocks apictl the first time, open **System Settings → Privacy & Security** and click **Allow Anyway**, or run `xattr -d com.apple.quarantine <APICTL_DIR>/apictl`.

3. Initialise apictl by running it once. The first run creates the apictl configuration folder `~/.wso2apictl` (Windows: `%USERPROFILE%\.wso2apictl`), where environments and exported APIs are stored.

   ```
   apictl version
   apictl get envs
   ```

   Expected output: the version (4.7.x) and an environment table. On a new installation the table has only the column headings:

   ```
   Version: 4.7.2
   NAME   API MANAGER ENDPOINT   REGISTRATION ENDPOINT   TOKEN ENDPOINT   ...
   ```

Part 2 – Add the Environment and Log In

4. Point apictl to the API Manager in which you want to deploy APIs by adding an environment named dev:

   ```
   apictl add env dev --apim https://<APIM_HOST>:<HTTPS_PORT>
   ```

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Example:

```
apictl add env dev --apim https://localhost:9443
```

Expected output:

```
Successfully added environment 'dev'
```

5. Log in to the environment. If you leave out `-p`, apictl asks for the password, which avoids typing it on the command line:

```
apictl login dev -u admin -k
```

Expected output:

```
Logged into APIM in dev environment
```

apictl also warns that the credentials are stored in plain text in its `keys.json` file.

Part 3 – Create the API

6. Initialise an API project from the Swagger Petstore OpenAPI definition. This creates a folder named `PetstoreAPI` in the current folder:

```
apictl init PetstoreAPI --oas https://petstore.swagger.io/v2/swagger.json
```

7. Open `PetstoreAPI/api.yaml` in an editor (for example VS Code). It is in the root of the project folder; there is no Meta-information folder in API Controller 4.x. The API name (`SwaggerPetstore`), context (`/v2`) and version (`1.0.7`) come from the OpenAPI file.

8. Change these values in api.yaml and save the file. Keep the spaces exactly as shown; YAML uses indentation.

```
lifeCycleStatus: PUBLISHED
endpointConfig:
  production_endpoints:
    url: https://petstore.swagger.io/v2
  sandbox_endpoints:
    url: https://petstore.swagger.io/v2
```

Part 4 – Publish the API

9. Push the API to API Manager. The project's `deployment_environments.yaml` deploys it to the **Default** gateway, and `lifeCycleStatus: PUBLISHED` publishes it:

```
apictl import api --file ./PetstoreAPI --environment dev -k --rotate-revision
```

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10. Check the result. SwaggerPetstore 1.0.7 is listed with the status PUBLISHED. You can also open the Publisher or the Developer Portal.

```
apictl get apis -e dev -k
```

11. Optional: invoke the API. The Petstore OpenAPI file protects its resources with the OAuth scopes `read:pets` and `write:pets`, so select `read:pets` when you generate the access token in the Developer Portal (otherwise the gateway returns 403). Then call:

macOS / Linux

```
curl -k -H "Authorization: Bearer <ACCESS_TOKEN>" \
  "https://<APIM_HOST>:<GW_HTTPS_PORT>/v2/1.0.7/pet/findByStatus?status=available"
```

Example – macOS / Linux:

```
curl -k -H "Authorization: Bearer eyJ4NXQiOiJNell4.eyJzdWIiOiJhZG1pbiJ9.ZXhhbXBsZQ" \
  "https://localhost:8243/v2/1.0.7/pet/findByStatus?status=available"
```

Windows (Command Prompt)

```
curl -k -H "Authorization: Bearer <ACCESS_TOKEN>" ^
  "https://<APIM_HOST>:<GW_HTTPS_PORT>/v2/1.0.7/pet/findByStatus?status=available"
```

Example – Windows (Command Prompt):

```
curl -k -H "Authorization: Bearer eyJ4NXQiOiJNell4.eyJzdWIiOiJhZG1pbiJ9.ZXhhbXBsZQ" ^
  "https://localhost:8243/v2/1.0.7/pet/findByStatus?status=available"
```

Expected Outcome

The SwaggerPetstore API has been created from an OpenAPI definition with the API Controller, deployed and published, and can be found in the Developer Portal.

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